

Koketse Sethosa Exercise 4

Exercise 4 : Date Functions

Question 1

```
SELECT order_id,  
       customer_id,  
       order_date,  
       DAYNAME(order_dateorder_date) AS DAY day_name  
FROM orders;
```

Output Table

order_id	customer_id	order_date	day_name
1001	101	2026-05-01	FRIDAY
1002	102	2026-05-02	SATURDAY
1003	103	2026-05-03	SUNDAY
1004	104	2026-05-04	MONDAY
1005	105	2026-05-05	TUESDAY

```

2) SELECT customer-id,
           customer-name,
           signup-date,
           MONTHNAME(signup-date) AS
           signup-monthname
FROM customer-signups;

```

Output Table

customer-id	customer-name	signup-date	signup-month-name
201	John	2026-01-15	Jan
202	Mary	2026-02-20	Feb
203	Peter	2026-03-05	Mar
204	Sarah	2026-04-18	Apr
2045	Thabo	2026-05-30	May

3) SELECT sale_id,
product_name,
sale_date,
MONTH(sale_date) AS sale_month
FROM sales;

sale_id	product_name	sale_date	sale_month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

Out

Question 4

```
SELECT transaction_id,  
       customer_id,  
       transaction_date,  
       YEAR(transaction_date) AS transaction_year  
FROM transactions;
```

Output Table

transaction_id	customer_id	transaction_date	transaction_year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

Question 5

```
SELECT delivery_id ,  
       customer_id ,  
       delivery_date ,  
       DAY (delivery_date) AS delivery_date  
FROM deliveries ;
```

Output Table

delivery_id	customer_id	delivery_date	day_of_month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

Question 6

```
SELECT employee-id,  
       employee-name,  
       department,  
       CURRENT_DATE() AS today-date  
FROM   employees;
```

Output Table

employee-id	employee-name	department	today-date
301	Nandi	Sale	2026-07-21
302	Brian	IT	2026-07-21
303	Lerato	Finance	2026-07-21
304	Sipho	HR	2026-07-21
305	Aisha	Marketing	2026-07-21

Question 7

```
SELECT order-id,  
       customer-id,  
       order-date-text,  
       TO_DATE(order-date-text) AS order-date  
FROM online-orders;
```

Output Table

order-id	customer-id	order-date-text	order-date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-0 ⁴ - 10	2026-04-10
4005	105	2026-05-05	2026-05-05

Question 8

```

SELECT payment-id
       customer-id
       payment-date
       formatted-payment-date TO-CHAR(payment-date)
       AS formatted-payment-date
FROM   payment-dates;
  
```

Output Table

payment-id	customer-id	payment-date	formatted-payment-date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-26

```

SELECT payment-id,
       customer-id,
       payment-date,
       TO-CHAR(payment-date, 'yyyy-MM-dd')
       AS formatted-payment-date
FROM   payments-dates;
  
```

Question 9

```
SELECT customer_id,  
       customer_name,  
       last_purchase_date,  
       DATEDIFF(DAY, last_purchase_date,  
                CURRENT_DATE())  
FROM customer_purchases;
```

Output Table

customer_id	customer_name	last_purchase_date	days_since_last_purchase
501	John	2026-05-01	84
502	Mary	2026-05-10	75
503	Peter	2026-05-15	70
504	Sarah	2026-05-20	65
505	Thabo	2026-05-25	60

Question 10

~~order-id~~ ~~customer-id~~ ~~order-date~~
SELECT order-id,
customer-id,
order-date,
~~expected~~ DATEADD(day, 7, order-date)
AS expected-delivery-date
FROM shipping-orders;

Output Tax

order-id	customer-id	order-date	expected-delivery-date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

Question 11

```
SELECT booking-id,
       customer-id,
       booking-date,
       DAY(booking-date) AS booking-year,
       MONTH(booking-date) AS booking-month,
       YEAR(booking-date) AS booking-year
FROM bookings;
```

Output Table

booking-id	customer-id	booking-date	booking-year	booking month	booking-day
Boo 1	101	2026-01-12	2026	1	12
Boo 2	102	2026-02-18	2026	2	18
Boo 3	103	2026-03-22	2026	3	22
Boo 4	104	2026-04-09	2026	4	9
Boo 5	105	2026-05-27	2026	5	27

Question 12

```
SELECT order-id,  
       customer-id,  
       order-date,  
       order-year YEAR(order-date) AS order-year,  
       amount  
FROM yearly-orders  
WHERE YEAR(order-date) = 2026;
```

Output Table

order-id	customer-id	order-date	order-year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000 1500

Question 13

```
SELECT order-id,  
       customer-id,  
       order-date,  
       MONTH(order-date) AS order-month,  
       amount  
FROM monthly-orders  
WHERE MONTH(order-date) = 3;
```

Output Table

order-id	customer-id	order-date	order-month	amount
8003	103	2026-03-10	3	800
8004	104	2026-03-25	3	1500
8005	105			

Question 14

```
SELECT    subscription_id ,  
          customer_id ,  
          start_date ,  
          LAST-DAY (start_date) AS month_end_date  
FROM      subscriptions ;
```

Output Table :

sub subscription_id	customer_id	start_date	month_end_date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-31
SUB004	104	2026-04-25	2026-04-30
SUB005	105	2026-05-30	2026-05-31

Question 15

```
SELECT  send-id ,  
        customer-id ,  
        send-date ,  
        month-start-date DATE_TRUNC ('month',  
        send-date ) AS month-start-date  
FROM    campaign-sends ;
```

Output Table.

send-id	customer-id	send-date	month-start-date
C001	101	2026-01-12	2026-01-01T00:00:00.000
C002	102	2026-02-18	2026-02-01T00:00:00.000
C003	103	2026-03-22	2026-03-01T00:00:00.000
C004	104	2026-04-09	2026-04-01T00:00:00.000
C005	105	2026-05-27	2026-05-01T00:00:00.000

Question 156

```
SELECT invoice-id,  
       customer-id,  
       invoice-date,  
       CONCAT(  
           DATENAME(MONTH, invoice-date), ' ',  
           YEAR(invoice-date)) AS invoice-month-year.  
FROM invoice-dates;
```

invoice-id	customer-id	invoice-date	invoice-month-year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

Question 17

```
SELECT customer-id,  
       customer-name,  
       date-of-birth,  
       DATEDIFF(YEAR, date-of-birth, (CURRENT-DATE))  
       AS customer-age,  
FROM customer-birthdays;
```

Output Table

customer-id	customer-name	date-of-birth	customer-age
901	John	1994 ⁸ -05-10	28
902	Mary	1990-08-20	36
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	41
905	Thabo	2000-07-30	26

Question 18

```
SELECT order_id,  
       customer_id,  
       order_date,  
       DAYNAME(order_date) AS Dayname - Name,  
       CASE  
         WHEN Day-Name = 'Sun' THEN 'Weekend'  
         WHEN Day-Name = 'Sat' THEN 'Weekend'  
         ELSE 'Weekday'  
       END AS Day-type  
FROM   weekend-orders;
```

Output Table

order_id	customer_id	order_date	day-name	day-type
9001	101	2026-05-01	Fri	Weekday
9002	102	2026-05-02	Sat	Weekend
9003	103	2026-05-03	Sun	Weekend
9004	104	2026-05-04	Mon	Weekday
9005	105	2026-05-05	Tue	Weekday

Question 19

```
SELECT transaction_id  
customer_id  
transaction_date  
QUARTER(transaction_date) AS transaction-quarter,  
amount  
FROM quarterly_transactions;
```

Output Table:

transaction_id	customer_id	transaction_date	transaction-quarter	amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-20	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

Question 20

```

SELECT order-id,
       customer-id,
       order-date,
       DATEDIFF(DAY, order-date, CURRENT_DATE())
       AS days-since-order,
       amount
FROM recent-orders;
WHERE DATEDIFF(DAY, order-date, CURRENT_DATE())
      AS > 30 ;
  
```

Output Table

order-id	customer-id	order-date	days-since-order	amount
R001	101	2026-04-01	119	500
R002	102	2026-04-15	105	1200
R003	103	2026-04-01	89	500
R004	104	2026-05-10	80	1500
R005	105	2026-05-25	65	2000

Bonus Challenge

```
SELECT customer_id,  
       customer_name,  
       last_purchase_date,  
       days-since-last-purchase DATEDIFF(DAY,  
       last_purchase_date, CURRENT_DATE())  
       AS days-since-last-purchase,  
       CASE  
         WHEN days-since-last-purchase <= 30  
           THEN 'Active Customer'  
         WHEN days-since-last-purchase  
           BETWEEN 31 AND 90  
           THEN 'At Risk Customer'  
         ELSE 'Inactive Customer'  
       END AS customer_status  
FROM customer_recency;
```

Output Table

customer-id	customer_name	last_purchase_date	days-since-last-purchase	customer-status
1001	John	2026-05-25	65	At Risk Customer
1002	Mary	2026-05-10	80	At Risk Customer
1003	Peter	2026-04-01	119	Inactive Customer
1004	Sarah	2026-02-15	164	Inactive Customer
1005	Thabo	2025-12-20	221	Inactive Customer